DAY 18 OF 30

Eventhouse and KQL

Where fast-moving data lives.
And the language built to query it.

Time series | Logs | Telemetry

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THE PROBLEM

A normal database chokes on event data.

1	**Volume**	Billions of rows a day
2	**Shape**	Semi-structured, always changing
3	**Questions**	"What happened in the last 5 minutes?"

Row-by-row databases were built for orders and customers, **not for a firehose**.

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THE SIMPLE VERSION

A search engine for events.

You do not read the whole internet to find a page. A search engine has already indexed it, so answers come back instantly.

An Eventhouse **indexes events as they arrive**, so queries over billions of rows return in seconds.

✓ **Stores**	Logs, telemetry, time series
✓ **Indexes**	Automatically, on arrival
✓ **Answers**	In seconds, at huge scale

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HOW IT WORKS

How an Eventhouse is organised

1. An **Eventhouse** holds one or more **KQL databases**
2. Events are **indexed and partitioned** as they land
3. You query with **KQL**, a pipe-based language
4. **Materialized views** keep common aggregates ready

KQL reads top to bottom

Take a table, filter it, summarise it. Each step is a pipe, like a recipe.

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FOR ENGINEERS

Settings that control cost and speed

Caching policy

Decides how much recent data is kept on fast storage. Hot data is fast, cold data is cheaper.

Retention policy

Decides how long data is kept at all. Set it per table, on purpose.

OneLake availability

Turn it on and the same data appears as Delta tables for Spark, SQL and Power BI.

Minimum capacity

An eventhouse can be kept always on, with a scheduled minimum size, to avoid slow first queries.

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THE HONEST PART

Eventhouse or Lakehouse?

EVENTHOUSE

Time-stamped events at high volume

Queries about the last minutes or hours

Free-text or semi-structured data

LAKEHOUSE

Large batch transformations

Data updated row by row

Classic star-schema reporting

> **The catch:** default retention and caching are rarely right for your data. Leave them alone and you either pay for history nobody queries or lose data someone needed.

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FROM REAL WORK

KQL on real projects

Two different uses of the same engine.

1. **A consumer delivery platform**: Eventhouse and KQL at sub-second latency
2. **Scale**: millions of daily active users
3. **A manufacturing plant**: image and sensor telemetry stored in KQL
4. **Purpose**: live monitoring of the factory floor

Very different businesses, **the same question: what is happening now?**

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SAVE THIS

Eventhouse in five lines

1. **Built for** events, logs and time series
2. **Indexes** everything on arrival
3. **KQL** filters and summarises in pipes
4. **Caching and retention** drive the cost
5. **OneLake availability** links it to the rest

Tomorrow: Dashboards and Activator. Seeing it live, and acting on it.

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